2. The Method declaration differs from the body as the declaration cites as the blueprint of the method, it gives it rules and a guide to what the body should somewhat look like as the method body is the code written in the body. An example of this interaction could be the declaration saying void, giving instructions to the body to not return the statement back to the main method.

3. An access modifier is the type of keyword used to change or define the access level of a method, variable, or class.

4. Visibility modifier

5
var1: local variable in the main method. It can only be accessed inside main.
var2: local variable in the first for loop in main. It can only be accessed inside that loop
var3: local variable in method1. It can be accessed throughout method1, including its for loop
var4: Local variable in the for loop inside method1. It can only be accessed inside that loop.

6.
a)
public static int GetVowels(String word)

b)
public static int extractDigit(int number)

c)
public static String insertString(String word, int number)

7.
a)
Using the concept called the method signature, just like humans use a full name to tell two people apart, a compiler looks at a specific combination of identifiers to uniquely track each method.

b)
Yes they can, this is known as method overloading

8.
a)
The return statement is used to send information back to whoever called the method, for example if the main method asked a method for the sum, it would find the sum then return it back to the main method.

b)
only one value is allowed, if more needed then you must make another method.

c)
The declaration of a method not returning a value will always be the type void, as this type does not allow returning at all, while returning methods have types such as int, string, and boolean.

9.
The main method is outside of the public class, this is not allowed as java doesn't like standalone methods that exist on their own.

11.

a) True

b) False

c) False

d) False

e) True

f) False

g) False

h) False

i) False

j) True

k) False

l) False